

Responses to Reviewers’ Comments

Manuscript: The Rise of Verbal Reinforcement Learning: A Survey (ACL ARR 2026 May, Submission 12040)

We sincerely thank the Area Chair and all four reviewers for their careful reading and constructive feedback. We have revised the manuscript to address every comment. All modifications are highlighted in **red** in the revised manuscript. Changes were made by direct replacement: superseded text was removed and the new or updated text appears in red. Page and line numbers below refer to the **marked, line-numbered version** of the revised manuscript. A companion “Summary of Differences” document reproduces the previous and updated text side by side for the major changes.

Reviewer pqAY

#	Reviewer comment	Our response
S	<i>Strengths: timely topic for the ACL community; unified framework for directions usually discussed separately; intuitive temporal framing; broad coverage; goes beyond listing prior work and identifies cross-cutting challenges.</i>	We sincerely thank the reviewer for this generous assessment. We have worked to preserve these strengths while tightening the scope as suggested below.
W1	<i>The scope of VRL could be more clearly delineated. The paper defines VRL broadly... it also makes the boundary with adjacent paradigms somewhat unclear... A more concrete discussion of what does not count as VRL, and how VRL differs from RLHF, instruction following, language-conditioned RL, and general LLM-agent feedback loops, would make the framing more precise.</i>	We fully agree, and we apologize for the imprecision. This comment is addressed as follows: the broad definition is replaced by two explicit inclusion criteria, (i) the linguistic structure of the feedback must be functionally consumed rather than collapsed to a scalar or preference bit before use, and (ii) the feedback must participate in an improvement loop (Section 1, page 2, lines 70–91). The same passage now states what does not count as VRL: plain instruction following (no improvement loop), vanilla RLHF/DPO (judgment reduced to a preference bit before learning; retained only as the scalar endpoint of the compression spectrum, see also Section 5.4, page 8, lines 617–627), and language-conditioned RL (language as policy input, not feedback). A boundary clarification for specification-time consumption appears at page 2, lines 89–93, and the abstract now announces the criteria (page 1, lines 8–11).

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W2	<i>The taxonomy would benefit from clearer rules for methods that span multiple pillars... Section 3.4 places reward-code generation under grounding... Section 4.4 treats experiential memory as inference-time deliberative feedback, but memory persists across episodes... explicit decision rules for such hybrid cases, or acknowledge these overlaps more systematically.</i>	We agree; this was underspecified. This comment is addressed as follows: Section 2 now states an explicit assignment rule, “a method is placed by what its linguistic signal modifies at the moment the signal takes effect” (page 3, lines 229–233), and a new hybrid-cases table (Table 1, page 4) records primary placements together with secondary roles instead of forcing single-cell membership. For the two cases the reviewer names: reward-code generation now carries the specification-time rationale and an explicit pointer to its downstream training role (Section 3.4, page 5, lines 368–372), and experiential memory is now classified by its moment of consumption with the cross-episode boundary recorded (Section 4.4, page 7, lines 467–472).
C1	<i>Please consider making terminology more consistent. For example, the paper appears to use both “grounded signal” and “grounding signal”; using one term throughout would improve clarity.</i>	Thank you for catching this; we apologize for the inconsistency. We have standardized on “grounding signal” and audited the full source: the revised text contains no remaining occurrence of “grounded signal” (the term now appears uniformly, e.g., Section 2 pillar definitions, page 3, and Section 3 throughout). We are additionally re-exporting the figures to ensure the embedded labels match the standardized term.
L1	<i>The societal impact discussion could be more explicit. Since VRL agents may act on natural-language feedback from humans, tools, or other models, it would be useful to discuss who may be harmed when feedback is incorrect, biased, adversarial, or poorly grounded... impacts on high-stakes agent applications such as coding assistants, robotics, clinical decision support, and education.</i>	We agree and thank the reviewer for the concrete framing. This comment is addressed in the expanded Limitations section (page 10, lines 813–831), which now names the failure modes (incorrect, biased, adversarial, or poorly grounded feedback) and discusses who is harmed in the high-stakes deployments the reviewer lists, complementing the existing Section 6 discussion of prompt injection, memory poisoning, and sycophancy.

Reviewer Q27s

#	Reviewer comment	Our response
S	<i>Strengths: the topic is timely and useful; the paper collects many relevant strands of work; I like the motivation for a survey in this area.</i>	We sincerely thank the reviewer for finding the direction exciting and for the detailed, actionable critique; the revision below follows it closely.
W1	<i>The definition of VRL is too broad. The paper defines VRL so broadly that many standard training paradigms could fit, including supervised fine-tuning, RLHF, DPO, and preference optimization. This makes the scope hard to distinguish from general LLM training/alignment.</i>	We agree, and we are sorry the original framing invited this reading. This comment is addressed as follows: the definition is replaced by two inclusion criteria (Section 1, page 2, lines 70–91) under which supervised fine-tuning on instructions and vanilla RLHF/DPO fall outside VRL; they are retained only as the scalar endpoint of the compression spectrum, stated explicitly where preference shaping is discussed (Section 5.4, page 8, lines 617–627). This materially narrows the scope and separates the survey from general LLM training and alignment.
W2	<i>The term VRL is already known in a narrower sense... introduced for self-reflective agents, but then adopts it much more broadly. This may confuse readers unless the authors more clearly justify the terminology and distinguish their usage.</i>	We agree. This comment is addressed by an explicit genealogy statement: Section 1 now says that we deliberately generalize the term introduced by Shinn et al. (2023) for self-reflective agents, and states the generalization up front to avoid conflation with the original narrower sense (page 2, lines 100–104).
W3	<i>The taxonomy feels forced. The three-pillar loop of grounding, feedback, and learning is intuitive as a conceptual picture, but no real method operates in this clean lifecycle. Presenting such a loop can misrepresent how verbal feedback actually works.</i>	We accept this criticism; the loop framing was misleading. This comment is addressed as follows: the revision removes the lifecycle implication and states that “the pillars classify individual signals, not stages of a pipeline: no method is claimed to traverse a grounding-deliberation-learning cycle” (Section 2, page 4, lines 242–247). The coding-agent example is reframed accordingly, as a system in which signals of all three kinds coexist.

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W4	<i>The survey should be more scoped... the paper effectively tries to cover most of modern LLM agent training and alignment. This breadth causes important omissions and makes the taxonomy less useful. A narrower focus, e.g., verbal feedback without model training, would make the contribution tighter.</i>	We agree that scoping was the core problem, and the new inclusion criteria (see W1) remove everything scalar-only from scope. We considered the reviewer’s specific suggestion of excluding model training entirely; we respectfully kept training-time methods that retain linguistic structure (feedback-conditioned training, critique-based distillation), because excluding them would sever the inference-time and training-time uses of the same signal. We hope the explicit fence, stated exclusions, and the boundary statement in Section 5.4 achieve the intended tightening.
W5	<i>The MDP framing is limiting... works for some agentic/RL settings, but not all language-agent or feedback-based methods naturally fit an MDP formulation especially considering the need for transition and reward functions.</i>	We agree in part, and we have clarified our intent. This comment is addressed by an explicit caveat in Section 2: MDP vocabulary serves as an organizing lens rather than a claim that every method admits an MDP formulation; many settings are partially observable, non-Markovian, or lack explicit transition and reward functions (pages 3–4, lines 235–241).
W6	<i>Search-guided deliberation does not clearly correspond to a feedback category in the same way as self-critique. It seems more like a decoding/search procedure than a distinct verbal-feedback mechanism.</i>	This is a fair point and we have revised accordingly. The subsection is renamed “Evaluator Feedback in Search” and now opens with the inclusion criterion: the verbal evaluator signal consumed during search is what qualifies, not the search procedure itself, and decoding strategies with no linguistic feedback fall outside VRL (Section 4.5, page 7, lines 487–493).

#	Reviewer comment	Our response
W7	<i>The paper would benefit from more synthesis or benchmarking. As written, the paper mostly describes methods. A benchmark, comparison framework, or more concrete analysis of when different verbal-feedback methods work would make the survey more valuable.</i>	We agree. This comment is addressed in three places: (a) a new evaluation-practice discussion recommending that feedback quality and feedback utilization be reported separately, with the relevant benchmarks (Section 6.2, page 9, lines 674–684); (b) a new practitioner decision table stating when each verbal-feedback mechanism is the right choice (Appendix B, Table 4, page 19); and (c) a named research agenda (Section 6.5, page 10, lines 758–773). Constructing a new benchmark is beyond a survey’s scope, but we believe the comparison framework and decision guidance directly address the comparability gap.
C1	<i>Reconsider whether “VRL” should be used this broadly.</i>	Addressed together with W1 and W2: the term is retained but its scope is now fenced by the two criteria, and the genealogy statement distinguishes our generalized usage from the original coinage.
C2	<i>Narrow the scope or clearly state what is excluded.</i>	Addressed together with W1 and W4: exclusions are now stated explicitly in Section 1 (page 2, lines 79–91), including instruction following, vanilla RLHF/DPO, and language-conditioned RL.

Reviewer 3gqM

#	Reviewer comment	Our response
S	<i>Strengths: relevant area; brings together strands often discussed separately; the specification/reasoning/reward-learning distinction is a useful organizing principle; may help readers see where language inserts into an RL pipeline.</i>	We sincerely thank the reviewer for these points and for the detailed guidance on what a survey should deliver; the revision follows it directly.

#	Reviewer comment	Our response
W1	<i>The paper needs a clearer comparison with prior surveys or related overview papers.</i>	We agree, and we apologize for the omission. This comment is addressed as follows: the introduction now positions the survey against prior overviews with citations (Section 1, page 3, lines 142–157), and a new comparison table covers twelve surveys and frameworks with columns for organizing axis, feedback types, lifecycle stages, and practitioner guidance (Appendix A, Table 3, page 19). The accompanying text states our differentiation: prior surveys organize by mechanism or agent capability, ours by the signal itself.
W2	<i>The taxonomy remains mostly high-level and descriptive.</i>	This comment is addressed by making the taxonomy operational: an explicit assignment rule (Section 2, page 3, lines 229–233), a hybrid-cases table showing the rule applied (Table 1, page 4), and a new contribution bullet committing the paper to applicability rather than description (Section 1, page 3, lines 177–180).
W3	<i>There is limited guidance on how practitioners should select among different language-feedback mechanisms.</i>	We agree this was missing. This comment is addressed by a new subsection, “Algorithmic Interactions and Practitioner Guidance” (Section 6.1, pages 8–9, lines 630–661), and a new decision table keyed to properties a practitioner can assess in advance, task verifiability, feedback latency and cost, and risk exposure, with representative methods per row (Appendix B, Table 4, page 19).

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W4	<i>The treatment of RL-specific issues is relatively shallow, especially regarding algorithm choice, credit assignment, and MDP assumptions.</i>	We agree and have deepened the treatment within the page budget. Section 6.1 now maps feedback forms to algorithm families (online policy gradients with generated rewards, offline preference objectives, supervised fine-tuning on filtered or conditioned text, and process reward models for step-level credit assignment) and discusses the MDP-level consequences the reviewer raises (pages 8–9, lines 630–661). Where formal results do not exist we say so explicitly, and the research agenda names credit assignment from free-form critiques as an open problem (Section 6.5, page 10).
C1	<i>The manuscript should explain more clearly what insight is gained from this grouping... provide practical guidance: when should language be used as a reward signal, when as state information, when for action-level guidance or reasoning?</i>	Addressed together with W3: Appendix B, Table 4 answers exactly this question row by row (language as reward, as state or task grounding, as deliberative feedback, as memory, and as training signal), each with its conditions, dominant risk, and representative methods.
C2	<i>The RL side needs more depth: language-based reward design, state abstraction, and policy generation may require different assumptions and work with different RL families... discuss the consequences of using language within the MDP formulation.</i>	Addressed together with W4: the new Section 6.1 discusses precisely these three consequence classes, mis-specification and hacking risks for language-specified rewards, information loss and aliasing for language-based state, and exploration effects for LLM-prior action generation (page 9), and the MDP-as-lens caveat clarifies the formal assumptions (Section 2, pages 3–4, lines 235–241).
C3	<i>The scope should be made more precise: the chosen terminology may suggest a broader set of feedback forms, including spoken, visual, or other multimodal signals. Either narrow the framing to language-centered feedback or expand to these modalities.</i>	We agree and have chosen to state the scope precisely: the feedback signal is natural-language text, while the agent’s observations may be non-textual, as in embodied settings (Section 1, page 2, lines 92–97). A fuller treatment of embodied and multimodal settings is planned for an extended version.

#	Reviewer comment	Our response
L1	<i>The paper should more directly discuss the limitations of language-based feedback in RL... reward specification failures, unreliable language interpretations, weak grounding, and challenges in transferring language-guided policies to real-world settings.</i>	We thank the reviewer for the precise list. This comment is addressed in the expanded Limitations section, which now names all four failure modes verbatim and connects them to the mitigation discussions in Section 6 (page 10, lines 813–831).

Reviewer bnae

#	Reviewer comment	Our response
S	<i>Strengths: timely and relevant topic; the proposed categorization provides a useful starting point; brings together several recent directions that are often discussed separately.</i>	We sincerely thank the reviewer for the frank and constructive assessment; it shaped the largest changes in this revision.
W1	<i>The contribution beyond existing literature is not sufficiently clear. The paper reads primarily as a synthesis of prior work, but it does not yet establish what new conceptual framework, technical insight, or research agenda it contributes.</i>	We accept this criticism. This comment is addressed by stating four checkable contributions in the introduction: (i) explicit inclusion criteria for the paradigm, (ii) decision rules with a hybrid-cases table, (iii) a signal-centric organization distinguishing this survey from mechanism- and agent-centric overviews, and (iv) practitioner guidance plus a research agenda (Section 1, page 3, lines 164–184). The comparison table (Appendix A, Table 3, page 19) makes the differentiation claim verifiable rather than asserted.
W2	<i>The paper lacks a clear call to action. It is not obvious how the proposed categorization helps researchers or practitioners define better problems, design better RL systems, or identify concrete future research directions.</i>	We agree this was missing. This comment is addressed by (a) the research agenda in Section 6.5, which names five open problems, sample-efficiency conditions, optimality under language-shaped rewards, credit assignment from critiques, feedback provenance and robustness, and standardized evaluation (page 10, lines 758–773), and (b) the practitioner decision table, which turns the taxonomy into design guidance (Appendix B, Table 4, page 19).

#	Reviewer comment	Our response
W3	<i>The paper would benefit from deeper discussion of how different RL algorithms interact with language-based feedback. A survey of this area should clarify which algorithmic settings are more suitable for different types of language feedback.</i>	We agree. This comment is addressed by the new Section 6.1, which pairs each feedback form with its optimization family and states the assumptions each pairing carries (pages 8–9, lines 630–661). Where the honest answer is that no formal result exists, the text says so and routes the question to the research agenda (Section 6.5).
W4	<i>The terminology appears too broad. The use of “verbal” may suggest a wider multimodal scope, including audio, video, or other communication modalities, while the paper seems mainly focused on language-based feedback.</i>	We agree and have stated the modality scope explicitly: the feedback signal is natural-language text; observations may be non-textual (Section 1, page 2, lines 92–97). The genealogy note (page 2, lines 100–104) additionally records that we generalize Shinn et al.’s narrower coinage deliberately.
C1	<i>The paper would be substantially stronger if it more clearly explained how it differs from existing surveys and related literature. . . it is not clear whether the taxonomy leads to new research questions, design principles, or new understanding.</i>	Addressed together with W1 and W2: the comparison table (Appendix A), the four stated contributions (Section 1), the design guidance (Appendix B), and the research agenda (Section 6.5) are all new in this revision and directly answer the three uses the reviewer names.
C2	<i>I encourage the authors to develop a more rigorous framework for analyzing language in RL. For example, when language replaces or augments reward signals, discuss how this affects the underlying MDP formulation, policy optimality, and temporal credit assignment. Similarly, when language is used as state information, examine whether such representations preserve the assumptions needed for RL.</i>	We thank the reviewer for this direction. Within the conference page budget, Section 6.1 now discusses the consequences of language inside the decision problem (reward mis-specification and hacking, state information loss and aliasing, exploration under LLM priors; page 9) and Section 6.5 names policy optimality and temporal credit assignment as open problems (page 10). We are honest that a full formal framework does not yet exist in the field; developing one is the first item of the stated agenda, and a comprehensive treatment is planned for an extended version.

#	Reviewer comment	Our response
C3	<i>The paper should also include a stronger algorithmic perspective. Different RL algorithms may benefit from language feedback in different ways, and it would be valuable to discuss which methods are appropriate under which conditions.</i>	This comment is addressed together with W3: the new Section 6.1 maps each feedback form to its optimization family with the conditions under which each pairing is appropriate (pages 8–9, lines 630–661), and Appendix B, Table 4 (page 19) states the per-mechanism conditions and risks.
L1	<i>The paper should more directly discuss the limitations of language-based feedback in RL, especially in theoretical and practical terms... possible failures in reward specification, unreliable language interpretations, weak grounding, and challenges in transferring language-guided policies to real-world settings.</i>	Addressed identically to Reviewer 3gqM’s request: the expanded Limitations section names all four failure modes and their societal-impact consequences (page 10, lines 813–831).